1. Item 7 (Most relevant)

> **Manipulate numbers, algebraic expressions and equations**

How it is reflected:

- The problem requires direct manipulation of the algebraic expression $400000y - 3200000$.
- Students must solve the inequality:

$$400000y - 3200000 \geq 0 \Rightarrow y \geq 8$$

👉 The core mathematical demand is **algebraic manipulation and solving equations/inequalities**.

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2. Item 12

> **Use a multi-step procedure leading to a mathematical solution**

How it is reflected:

- Formulating an inequality based on the condition "financial gain ≥ 0"
- Rearranging and simplifying the expression
- Solving for y
- Interpreting the solution and selecting the smallest valid integer

👉 The solution requires a **clear sequence of logical steps**, not a single operation.

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3. Item 5

> **Understand and utilise constructs based on definitions, rules and formal systems**

How it is reflected:

- Understanding that "covering the construction cost" means the financial gain is zero or positive
- Correctly applying the rules for solving linear inequalities
- Using standard algebraic conventions and notation

👉 This reflects correct use of **formal mathematical definitions and rules**.

4. Item 13

> **Use an understanding of the context to guide or expedite the mathematical solving process**

How it is reflected:

- Recognising that y represents years and therefore must be a non-negative integer
- Interpreting $y = 8$ as "at least 8 years"
- Connecting the algebraic result to the real-world context of operating a wind power station

👉 Contextual understanding is essential to arriving at a **meaningful and realistic answer**.

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5. Item 1

> **Perform a simple calculation**

How it is reflected:

- Basic arithmetic operations such as addition, subtraction, and division
- Although the calculations are straightforward, they are essential to the solution

👉 This represents fundamental computational skills.

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6. Item 8

> **Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**

How it is reflected:

- A complete solution requires clearly showing the inequality setup, intermediate steps, and the final conclusion
- This is particularly relevant in written or extended-response formats

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7. Item 2

> **Select an appropriate strategy from a list**

How it is reflected:

- Choosing to solve an inequality rather than testing each answer option individually
- Using an algebraic approach as the most efficient strategy

👉 The strategic choice is present but limited, so its relevance is moderate.

① 第 7 项（最相关）

> **Manipulate numbers, algebraic expressions and equations**

核心体现：

- 直接对代数式 $400000y - 3200000$ 进行操作
- 解不等式：

$$400000y - 3200000 \geq 0 \Rightarrow y \geq 8$$

👉 本题的数学核心就是**代数运算与方程/不等式求解**

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② 第 12 项

> **Use a multi-step procedure leading to a mathematical solution**

体现方式：

- 建立不等式（收益 ≥ 0）
- 化简、移项
- 求解
- 结合现实情境取最小整数年数

👉 解题不是一步完成，而是**清晰的多步骤推理过程**

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③ 第 5 项

> **Understand and utilise constructs based on definitions, rules and formal systems**

体现方式：

- 理解"收回成本"= 财政收益为 0 或正
- 正确理解线性函数的结构和规则
- 使用一元一次不等式的规范解法

👉 这是对**数学定义和规则的正确运用**

④ 第 13 项

> **Use an understanding of the context to guide the mathematical solving process**

体现方式：

- 理解 y 代表"年数"，不能是负数或小数
- 解得 $y = 8$ 后，理解"至少需要 8 年"
- 将数学结果与现实问题（风电站运营）相联系

👉 情境理解直接影响最终答案的合理性

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⑤ 第 1 项

> **Perform a simple calculation**

体现方式：

- 基本的移项、除法计算
- 运算难度低，但不可缺失

👉 属于基础计算能力的直接体现

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⑥ 第 8 项

> **Articulate a solution, showing intermediate mathematical results**

体现方式：

- 在完整解答中，需要展示：
  - 不等式建立
  - 中间计算步骤
  - 最终结论

👉 尤其在考试或书面表达中非常重要

⑦ 第 2 项

> **Select an appropriate strategy from a list**

体现方式：

- 选择"设收益 ≥ 0"而不是代值试算
- 在选项题中，策略是"先解方程，再对照选项"

👉 但策略选择空间较小，因此相关性中等